

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) — Semester II Examination, 2023-24

Paper No. MTB-201: Calculus-II

Time: 3 Hours

Maximum Marks: 70

Ref. No.: 074896

*Instructions: Solve **any five** questions. Question 1 is compulsory.***Question 1.** (Compulsory — 2 marks each.)

- (a) Let $f(x, y) = \frac{xy}{\sqrt{x^2 + y^2}}$ for $(x, y) \neq (0, 0)$. Check whether $\lim_{(x, y) \rightarrow (0, 0)} f(x, y)$ exists.
- (b) Let $f(x, y) = \begin{cases} \frac{x^2 y^2}{x^2 + y^2} & (x, y) \neq (0, 0) \\ 0 & x = y = 0 \end{cases}$. Find f_x and f_y at the origin and test continuity of f_x at the origin.
- (c) If $u = \sin(x^2 + y^2)$ where $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, find $\frac{du}{dx}$.
- (d) Define a homogeneous function of degree n in four variables x, y, z, w and give an example.
- (e) Let U be a non-empty open set in $\mathbb{R}^2$ and $f : U \rightarrow \mathbb{R}$. Give the definition of the derivative of f at $(a, b) \in U$. Find the derivative of $f(x, y) = x^2 y + xy^2$ at $(-1, -1)$.
- (f) Change the order of integration in $\int_0^\infty dy \int_y^\infty e^{-x^2} dx$ and evaluate it.
- (g) Find the volume of the solid bounded by $x^2 + y^2 = 1$ and the planes $y + z = 1$ and $z = 0$.

Question 2.

- (a) Let $f, g : D \rightarrow \mathbb{R}$ with $\lim_{(x, y) \rightarrow (a, b)} f(x, y) = \ell$ and $\lim_{(x, y) \rightarrow (a, b)} g(x, y) = m$. Show that $\lim_{(x, y) \rightarrow (a, b)} f(x, y)g(x, y) = \ell m$. [5]
- (b) State and prove Young's theorem for a function of two variables. Verify with an example. [9]

Question 3.

- (a) Let $f : D \rightarrow \mathbb{R}$ be differentiable at $(a, b) \in D$. Show that f is continuous at (a, b) . Give an example of $f(x, y)$ continuous at $(0, 0)$ but not differentiable there. [6]

- (b) Let $C \subseteq \mathbb{R}^2$ with $(a, b) \in \text{int}(C)$ and $f : C \rightarrow \mathbb{R}$. Suppose $\exists r > 0$ such that (i) $B_r((a, b)) \subseteq C$; (ii) $f_x(x, y)$ exists $\forall (x, y) \in B_r((a, b))$ and f_x is continuous at (a, b) ; (iii) f_y exists at (a, b) . Show that f is differentiable at (a, b) . [8]
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Question 4.

- (a) State and prove Euler's theorem for homogeneous functions of degree n in x, y, z . Verify for a homogeneous function of degree 3. [5]
- (b) Prove that $\frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2}$ is invariant under change of rectangular axes. [5]
- (c) Show that $u = x + 3y + 2z$, $v = 3x + 4y - 2z$, $w = 11x + 18y - 2z$ are not independent and find the relation among them. [4]
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Question 5.

- (a) State and prove Taylor's theorem for expansion of $f(x, y)$ about the point (a, b) . [7]
- (b) Find the minimum value of $x^2 + y^2 + z^2$ subject to $xy + yz + zx = 12$. [7]
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Question 6.

- (a) Evaluate $\iint (x + y)^2 dx dy$ over the area bounded by $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$. [7]
- (b) Evaluate $\iint r \sin \theta dr d\theta$ over the area of the cardioid $r = a(1 + \cos \theta)$ above the initial line. [7]
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Question 7.

- (a) Establish the relation between Beta and Gamma functions. [7]
- (b) Evaluate $\iiint e^{x+y+z}(1-x-y-z) dx dy dz$ over the tetrahedron $x+y+z \leq 1$, $x, y, z \geq 0$. [7]
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